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Waste-to-feed bioconversion using *Hermetia illucens* Larvae: Current insights and prospects

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ABSTRACT

The rapid growth of the global population, urbanization, and intensive livestock production has led to a surge in organic waste generation and an increasing demand for sustainable feed sources. Black soldier fly (*Hermetia illucens*, BSF) larvae have emerged as a promising solution for converting organic waste into valuable biomass. These larvae efficiently degrade a variety of substrates—such as food waste, agricultural by-products, and animal manure—transforming them into protein- and fat-rich biomass suitable for animal feed, biofertilizers, and other applications. This review summarizes recent developments in the use of BSF larvae for waste-to-feed bioconversion. It explores larval growth performance, feed substrate characteristics, biomass nutritional composition, processing advancements, and safety considerations. BSF larvae consistently demonstrate high conversion rates and generate biomass rich in nutrients and bioactive compounds, making them ideal feed ingredients for poultry, pigs, aquaculture species, and pets. Research has advanced in optimizing rearing conditions and postprocessing methods to improve efficiency and product quality. Nonetheless, substrate standardization, microbial risk control, cost reduction, and the development of scalable systems for industrial adoption remain challenges. The review also addresses regulatory frameworks and social acceptance issues that influence BSF-based product commercialization. *Hermetia illucens* larvae offer an environmentally sound and innovative approach to organic waste management while contributing to sustainable protein production. Their successful integration into Circular Economy Models depends on further technological improvements, supportive policies, and stakeholder collaboration. With continued efforts from researchers, industry, and regulators, BSF bioconversion can revolutionize waste treatment and animal nutrition, thereby supporting a more resilient and resource-efficient food system.

Keywords: Bioconversion, *Hermetia illucens*, Insect protein, Organic waste, Sustainable feed.

Introduction

The growing problem of organic waste accumulation resulting from agricultural, food, and industrial activities is facing the modern world (Thyberg and Tonjes, 2016). Agriculture—including both crop and livestock production—generates millions of tons of waste every year, such as crop residues (tops, husks, ungraded produce), processing by-products (oil cakes, pomace), manure, and poultry litter (Westendorf, 2000). According to the Food and Agriculture Organization of the United Nations (FAO), approximately 1.3 billion tons of food waste are produced annually worldwide, accounting for one-third of all food produced (FAO, 2013). In the absence of an effective disposal system, these wastes, including straw, leaves, and food processing residues, create serious environmental, economic, and social problems.

The accumulation of organic waste leads to soil, water, and air pollution, as well as greenhouse gas emissions such as methane and carbon dioxide, which intensify the greenhouse effect and contribute to climate change

(Levis and Barlaz, 2011). Improper manure and plant residue storage results in methane and ammonia emissions, which, according to Stephen M. According to Ogle *et al.* (2014), agriculture accounts for up to 20% of global greenhouse gas emissions. Nitrate and phosphate leaching from manure into groundwater causes eutrophication of water bodies (Carpenter, 1998). Moreover, long-term storage of untreated waste promotes the release of toxic substances, such as hydrogen sulfide, and the proliferation of pathogenic microorganisms, posing a threat to the health of local communities (Zhang *et al.*, 2021). In many countries, especially developing ones, a significant share of waste is landfilled or incinerated due to a lack of infrastructure and technology, worsening environmental damage (Wilson *et al.*, 2006).

Population growth, which has reached 8 billion people (UN, 2022), urbanization, and the expansion of livestock farming, which also increases the demand for animal feed, drive the growth in organic waste volumes (Mottet *et al.*, 2017). Traditional feed

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production requires substantial resources such as land, water, and energy, placing additional pressure on the environment (FAO, 2021). Unused organic waste has high potential for conversion into biogas, compost, or organic fertilizers, emphasizing its value as a resource (Cherubini, 2010).

Finding effective methods for organic waste utilization has become a top priority amid a growing environmental crisis. One promising solution is bioconversion using black soldier fly larvae (*Hermetia illucens*), which not only significantly reduces waste volume but also produces valuable outputs, such as feed protein and organic fertilizers (Van Huis, 2013). This approach helps reduce landfill waste, produce sustainable alternatives to conventional feed, and lower the agro-industrial chain's carbon footprint (Gold *et al.*, 2021). The technology has been successfully implemented in various countries, and its scaling and automation have contributed to the widespread adoption of this model. The black soldier fly larvae (*Hermetia illucens*) represent a promising tool for organic waste bioconversion. This insect species is characterized by rapid growth, low maintenance requirements, and the ability to process a wide range of organic materials—from food and agricultural waste to manure and residues from the food processing industry—efficiently.

The bioconversion process involving *H. illucens* larvae is based on their natural ability to rapidly ingest and transform organic matter into valuable biomass rich in proteins, fats, and micronutrients. Simultaneously, the overall volume of waste is significantly reduced, the microbiological load is decreased, and environmental pollution is minimized.

Compared with traditional waste disposal methods (such as composting, incineration, and landfilling), the use of black soldier fly larvae offers several significant advantages. Research indicates that *H. illucens* larvae can reduce the mass of organic waste by 50%–80% within 10–14 days, while the substrate-to-larval biomass conversion rate (bioconversion rate) ranges from 15% to 25%, depending on the rearing conditions (Diener *et al.*, 2009; Surendra *et al.*, 2016).

Thus, *H. illucens* represents an environmentally friendly and highly efficient biological agent capable of playing a key role in the development of sustainable organic waste management systems and the provision of alternative sources of feed protein (Macwan *et al.*, 2024).

This article reviews the environmental and health impacts of organic waste accumulation and explores innovative processing methods using larvae of the black soldier fly (*Hermetia illucens*). This study aims to systematically analyze the research and applications of larvae as bioconversion agents for transforming organic waste into high-value feed. This review examines the biological characteristics, rearing technologies, nutritional composition, safety, and regulatory frameworks of larvae-derived products. Environmental

and economic benefits, such as waste reduction, resource efficiency, and greenhouse gas mitigation, are also evaluated. Finally, the study identifies knowledge gaps and outlines strategic directions for the further development of this sustainable technology in agriculture and the circular bioeconomy.

Hermetia illucens: Biology, Life Cycle, and Nutritional Value

Hermetia illucens: life cycle and ecological role

The black soldier fly (*Hermetia illucens*), a member of the Stratiomyidae family, is a dipteran insect species that plays a significant role in sustainable development due to its capacity to process organic waste and produce valuable products such as protein-rich biomass and biofertilizer (Diclaro and Kaufman, 2025). This species has attracted considerable attention in agriculture and the circular economy because its larvae efficiently convert food scraps, manure, and agricultural by-products into animal feed and fertilizer resources (Sheppard *et al.*, 1994; Van Huis, 2013).

Hermetia illucens stands out among insects because of its unique characteristics. Adult flies do not feed and are not vectors of pathogens, making them safe for use in agricultural systems (Wang and Shelomi, 2017). The primary value of this insect lies in its larvae, which can process up to 75% of organic waste, reduce its mass by 60%–72%, and transform it into a biomass rich in proteins (35%–40%) and fats (15%–25%) (Surendra *et al.*, 2016; Liu *et al.*, 2018). This makes the larvae a promising feed source for poultry, fish, and pigs, particularly in the context of the global protein shortage (Makkar *et al.*, 2014).

The ecological role of *H. illucens* is multifaceted. The larvae significantly reduce the volume of organic waste and help lower methane and other greenhouse gas emissions (Diener *et al.*, 2009). Composting organic waste with BSF larvae resulted in significant nutrient enrichment, with total phosphorus increasing up to 32.4% (compared to 10.3% in the control) and total potassium reaching a maximum mean increase of 77.1%, thereby emphasizing the strong fertilizing potential of the final compost (Sarpong *et al.*, 2019). Moreover, the larvae contribute to pathogen reduction by inactivating harmful bacteria such as *Salmonella spp.* and *Escherichia coli* (with a 99.9% reduction), thanks to their stable gut microbiota (Lalander *et al.*, 2013). They can also degrade microplastics, reducing the mass of polystyrene by 30%–40% within 30 days (Yang *et al.*, 2023).

From an economic perspective, the species is highly attractive: its larvae are used to produce protein meal and oil, which can substitute conventional feed ingredients such as soybean meal or fishmeal (Sprangers *et al.*, 2017).

The life cycle of *H. illucens* consists of four stages: egg, larva, pupa, and adult. Each stage has its own

Table 1. Key characteristics of the life cycle stages of *Hermetia illucens* (Tomberlin *et al.*, 2002; Rehman *et al.*, 2023).

Stage	Duration	Key characteristics	Environmental requirements	Role in bioconversion
Egg	4–6 days	500–900 eggs laid near organic matter; highly sensitive to temperature fluctuations	25°C–30°C; 60%–80% relative humidity	N/A (prepares for larval stage)
Larva	7–14 days (up to 20)	Main bioconversion stage; 5–6 instars; final dry mass ~43.5–44.0 mg; rich in protein and fat	27°C–30°C; substrate moisture 60%–80%; 2.5–5.0 larvae/cm ²	Converts organic waste into high-protein, high-lipid biomass
Pupa	7–14 days	Larvae stop feeding; metamorphosis occurs; sensitive to humidity and temperature	Stable temp/humidity needed for successful development	Prepares for adult emergence; no active bioconversion
Adult Fly	5–8 days	Does not feed; focused on mating and oviposition; not a pest or disease vector	Dry, protected areas for mating and egg-laying	Ensures reproduction; no role in bioconversion

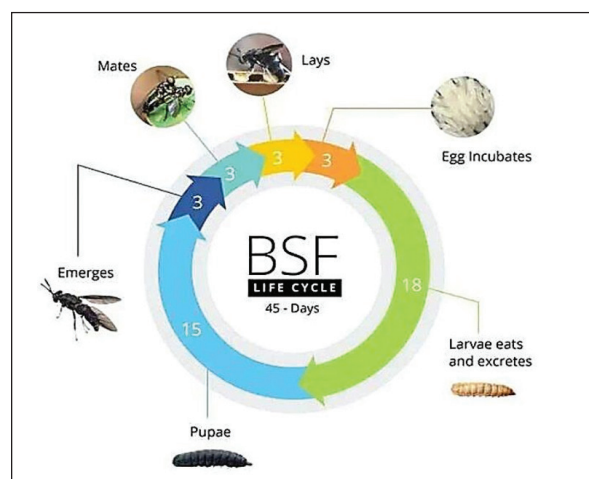


Fig. 1. Development stages of BSF (Source: Wong, 2020).

characteristics and plays a specific role in the process of bioconversion (Fig. 1, Table 1).

Egg stage: Female *Hermetia illucens* lay between 500 and 900 eggs in dry, protected areas near organic substrates. Incubation lasted 4–6 days at 25°C–30°C and relative humidity of 60%–80% (Tomberlin *et al.*, 2002). Extreme environmental conditions (temperatures 20°C or >40°C) significantly reduce egg viability (Chia *et al.*, 2018).

Larval Stage: This critical phase for bioconversion involves 5–6 instars over 7–14 days (up to 20 days under certain conditions), during which larvae reach a dry mass of approximately 43.5–44.0 mg. By day 14, the first prepupae appeared on all tested substrates, and by day 19, more than half had emerged, generally exceeding 210 mg per larva, except on poultry manure (165 mg), with the largest prepupae, averaging just

over 250 mg, developing on abattoir waste combined with fruits and vegetables (Lalander *et al.*, 2019). BSF larvae have a highly efficient digestive system that enables them to break down organic waste into valuable byproducts, such as protein-rich biomass and fertilizer. Their gut is equipped with digestive enzymes to efficiently process various organic materials, including amylase for carbohydrates, proteases for proteins, and lipases for lipids. They possess a highly efficient digestive system enriched with enzymes—amylases (85 U/g) and proteases (~100 U/g)—capable of breaking down carbohydrates, proteins, and lipids (Wang and Shelomi, 2017). Optimal rearing conditions include a temperature of 27°C–30°C, substrate moisture of 60%–80%, and larval density of 2.5–5.0 larvae/cm². Larvae process food waste, manure, and agricultural byproducts into high-value biomass containing 40%–45.2% protein and 20%–35% lipids, including antimicrobial lauric acid (Spranghers *et al.*, 2017).

High fiber content (>7%) slows larval growth, whereas substrate pretreatment (e.g., fermentation, grinding) can enhance bioconversion efficiency by 10%–15% (Liu *et al.*, 2018).

Pupal stage: At this stage, the larvae stop feeding and begin to pupate. The adult fly emerges from the pupa within 7–14 days. Environmental factors, such as temperature and humidity, influence the duration of this stage (Chia *et al.*, 2018).

Adult stage: Adult flies live for 5–8 days, do not feed, and focus exclusively on reproduction. They do not pose a threat as pests because they do not interact with food or crops (Sheppard *et al.*, 2002).

Hermetia illucens makes a significant contribution to the circular economy and sustainable agriculture. Its ability to process organic waste reduces the burden on landfills and mitigates greenhouse gas emissions,

while the production of biofertilizer and feed biomass supports environmentally friendly farming practices (Camperio *et al.*, 2025). The high nutritional value of the larvae—rich in essential amino acids (lysine, methionine, and threonine) and trace minerals (calcium, phosphorus, and magnesium)—makes them a competitive alternative to conventional feeds (Makkar *et al.*, 2014).

Functional characteristics of *H. illucens* Larvae in bioconversion processes

Hermetia illucens larvae can convert up to 50%–80% of the mass of organic waste—including food residues, pig manure, and agricultural by-products—within 2–4 weeks. Sheppard *et al.* (2002) highlighted their ability to effectively degrade pig manure, significantly reducing both its volume and environmental impact. According to Bruno *et al.* (2025), the powerful digestive system of larvae, rich in enzymes such as lipases and proteases, enables them to break down complex organic substrates. The larvae contain 40%–50% protein and 20%–35% fat on a dry matter basis, making them a valuable raw material for animal feed (Lu *et al.*, 2024).

Tahamtani *et al.* (2021) demonstrated that the inclusion of live larvae in the diets of laying hens had no negative effects on bird health or behavior and even increased feed palatability, confirming their nutritional value.

Thongpea *et al.* (2022) further emphasized that larval biomass is rich in essential amino acids (e.g., lysine and methionine) and fatty acids (notably lauric acid), making it suitable for use in both feed and biofuel production.

Larvae convert low-value substrates into nutrient-rich biomass (Rehman *et al.*, 2022). Bruno *et al.* (2025) emphasized that the larval digestive system—supported by symbiotic microorganisms—efficiently decomposes organic matter while accumulating proteins and lipids. Sheppard *et al.* (2002) noted that larvae reared on pig manure produce biomass suitable for feed, and Mou

et al. (2024) added that such biomass is already being used in aquaculture and livestock farming. According to Ogello *et al.* (2025) and Alagappan *et al.* (2025), the highest protein content was observed in larvae reared on mixed feed (49.2%), while the lowest was found in those fed on poultry manure (39.1%). This indicates that the substrate type has a significant impact on the larval nutritional profile (Fig. 2).

Hermetia illucens larvae develop from egg to prepupa within 14–28 days under optimal conditions (27°C–30°C temperature, 60%–70% humidity), ensuring high productivity. Sheppard *et al.* (2002) described cultivation methods that optimize this cycle, including the use of containers with controlled humidity. Bruno *et al.* (2025) and Tomberlin *et al.* (2015) reported that the physiological adaptations of larvae enhance growth on various substrates.

These larvae are highly tolerant to a broad range of substrates and environmental conditions (15°C–40°C and varying humidity levels). Bruno *et al.* (2025) explained that the digestive system of larvae adapts to diverse waste types due to a wide array of enzymes and gut microbiota (Nguyen *et al.*, 2015).

Alagappan *et al.* (2025) and Park *et al.* (2015) emphasized that this property makes bioconversion products safe for use in animal feed.

Bioconversion using BSF larvae significantly reduces landfill waste volume and greenhouse gas emissions. Číčková *et al.* (2015) added that larval frass (excrement) is a valuable organic fertilizer rich in nitrogen and phosphorus, contributing to sustainable agriculture.

Hermetia illucens larvae can process food waste, manure, grain residues, and even certain plastics (Fig. 3). While Sheppard *et al.* (2002) focused on pig manure, Bruno *et al.* (2025) highlighted the digestive capacity of larvae to handle complex substrates, including kitchen waste and industrial by-products. Yang *et al.* (2023) and their potential in aquaculture waste management.

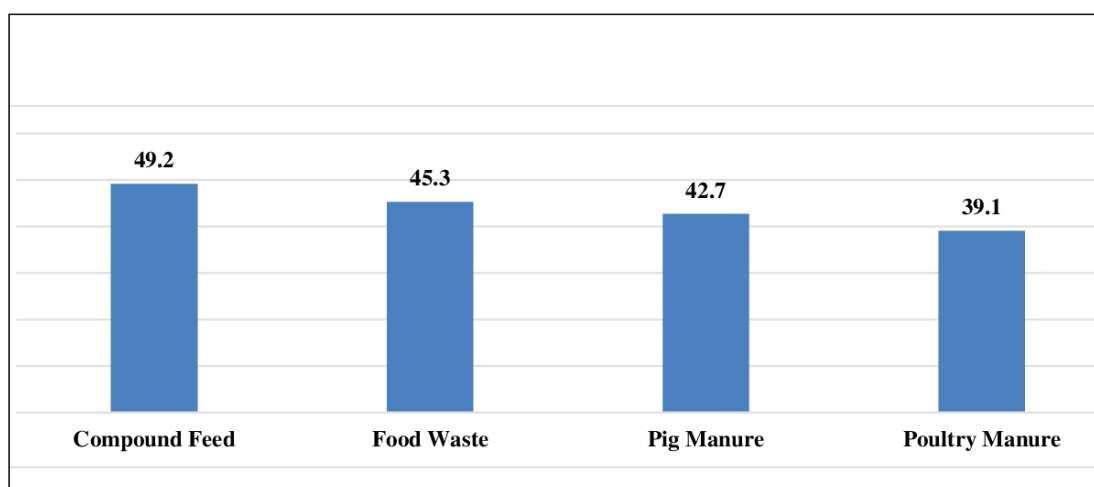


Fig. 2. Protein content in Larvae reared on different substrates, % (Ogello *et al.* (2025)).

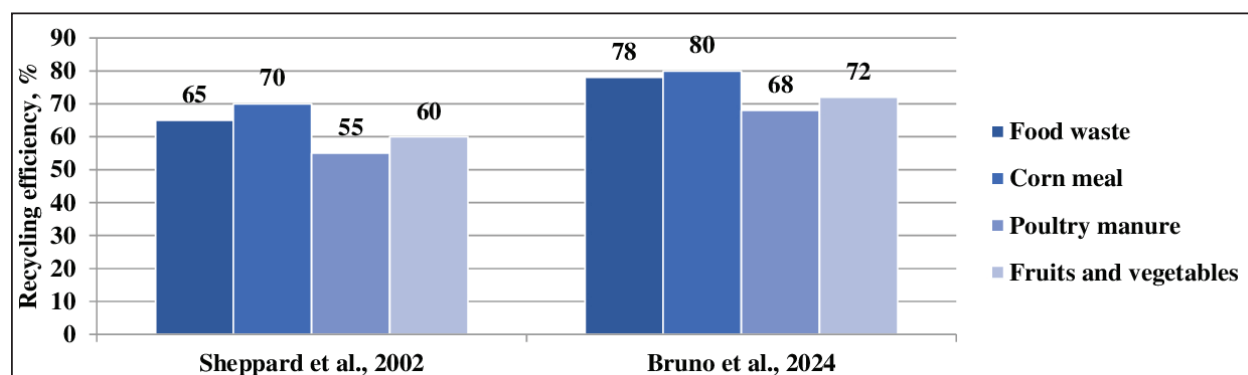


Fig. 3. Comparison of the efficiency of processing different wastes by BSF larvae, % (Sheppard *et al.*, 2002; Bruno *et al.*, 2024).

Table 2. Comparative nutritional profile of insect Larvae (Nekrasov *et al.*, 2019; Finke, 2002; Ewald *et al.*, 2020).

Nutrient	Black soldier fly (<i>Hermetia illucens</i>)	Mealworm (<i>Tenebrio molitor</i>)	Green bottle fly (<i>Lucilia sericata</i>)
Protein (%)	42–50	47–53	50–60
Fat (%)	30–35	20–30	15–25
Calcium (%)	5–8	0.1–0.3	1–2
Phosphorus (%)	1–2	0.7–1.0	0.8–1.5
Chitin (%)	5–8	2–4	3–5
Energy Value (MJ/kg)	18–22	18–20	17–19

Bioconversion of BSF larvae requires minimal infrastructure. Sheppard *et al.* (2002) described simple container-based cultivation methods, while Alagappan *et al.* (2025) (2024) emphasized the cost-effectiveness of this technology, making it suitable for both small- and large-scale operations. Larval biomass is used in animal feed, biofuels, cosmetics, and pharmaceuticals, while *frass* serves as a sustainable fertilizer. Tahamtani *et al.* (2021) confirmed that live larvae improve the diet of laying hens without affecting egg quality. Wang and Shelomi (2017) emphasized the high commercial potential of pig manure bioconversion products.

Comparative nutritional profile of feed insects

Insect larvae are a rich source of protein, with content varying depending on the insect species, developmental stage, and nutritional substrate (Table 2). According to various studies, the nutritional composition of *H. illucens* larvae fluctuates based on diet, developmental stage, and rearing conditions.

According to research, the crude protein content is as follows: black soldier fly (*Hermetia illucens*) larvae contain 42%–50% protein on a dry matter basis, increasing up to 65% after fat extraction (Nekrasov *et al.*, 2022); green bottle fly (*Lucilia sericata*) larvae can contain up to 60% protein when reared on optimized substrates, such as mixtures including camelina oil and mealworm (*Tenebrio molitor*) larvae, which contain approximately 47%–53% protein (Ooninex

and de Boer, 2012). Based on the studies by Nekrasov *et al.* (2019), Finke (2002), and Ewald *et al.* (2020), Table 3 summarizes the nutritional composition of three commonly used insect larvae species in animal feed: black soldier fly (*Hermetia illucens*), mealworm (*Tenebrio molitor*), and green bottle fly (*Lucilia sericata*).

The protein composition of insect larvae includes a wide range of essential amino acids—such as lysine, tryptophan, and methionine—whose profile is comparable to soybean meal and fishmeal, making insect larvae a promising alternative to conventional feed ingredients (Al-Qazzaz and Ismail, 2016). For example, dipteran larvae have higher lysine and tryptophan contents than crickets or locusts. *Hermetia illucens* larvae provide a valuable source of essential amino acids (Table 4; Makkar *et al.*, 2014).

Insect larvae also contain significant amounts of lipids, ranging from 15% to 40% of dry matter, with black soldier fly (*Hermetia illucens*) larvae containing up to 35% fat. Notably, their lipid profile includes high concentrations of lauric acid (up to 60% of total fat), which possesses strong antimicrobial properties (Ewald *et al.*, 2020) (Table 5). Lauric acid and specific antimicrobial peptides (e.g., cecropins and sarcotoxins) found in larvae have been shown to reduce pathogenic microbial load, thus potentially lowering the need for antibiotics in animal production (Borrelli *et al.*, 2021).

Table 3. Chemical composition of *Hermetia illucens* Larvae (average values on dry matter basis).

Parameter	Value (dry matter basis)	Source
Crude protein (%)	35–60	Makkar <i>et al.</i> , 2014; Henry <i>et al.</i> , 2015
Crude fat (%)	15–49	van Huis <i>et al.</i> , 2013; Surendra <i>et al.</i> , 2016
Ash (%)	7–15	Diener <i>et al.</i> , 2009
Fiber (%)	7–10	Wang & Shelomi, 2017
Calcium (%)	0.5–2.0	Barragán-Fonseca <i>et al.</i> , 2017
Phosphorus (%)	0.6–1.5	Schiavone <i>et al.</i> , 2017

Table 4. Amino acid profile of *Hermetia illucens* Larvae (Makkar *et al.*, 2014; Henry *et al.*, 2015; Schiavone *et al.*, 2017; Surendra *et al.*, 2016; van Huis *et al.*, 2013).

Amino acid	Content (g/100 g protein)	Source
Lysine	6.0–8.2	Makkar <i>et al.</i> , 2014
Methionine	1.4–2.1	Henry <i>et al.</i> , 2015
Threonine	4.0–4.8	Schiavone <i>et al.</i> , 2017
Valine	5.5–6.2	Surendra <i>et al.</i> , 2016
Isoleucine	4.2–5.0	van Huis <i>et al.</i> , 2013

Table 5. Fatty acid profile of *Hermetia illucens* Larvae (Ewald *et al.*, 2020).

Fatty acid	Content (%)	Characteristic
Lauric (C12:0)	28–55	Strong antimicrobial properties
Myristic (C14:0)	5–8	Saturated fatty acid
Oleic (C18:1)	10–20	Monounsaturated
Linoleic (C18:2, ω -6)	2–5	Polyunsaturated, omega-6
Alpha-linolenic (C18:3, ω -3)	0.2–1	Polyunsaturated, omega-3

Larval fats provide a high energy value of 18–22 MJ/kg and can be tailored to the nutritional needs of specific animal species by modifying the larval substrate. Insect larvae also supply important macro- and microelements, including calcium, phosphorus, magnesium, zinc, and B vitamins. Black soldier fly larvae contain up to 5%–8% calcium and 1%–2% phosphorus in their dry matter, which is higher than that in many conventional feed ingredients (Chobanova *et al.*, 2024).

The carbohydrate content in *H. illucens* larvae is relatively low, around 5%–10% of dry matter, and the chitin fraction accounts for approximately 4%–7% (ranging from 3.9% to 7.2%), which aligns with the estimated 5%–8% chitin as part of the total carbohydrate component in the larvae (Wang *et al.*, 2020; Lu *et al.*, 2022).

The energy value of insect larvae—18–22 MJ/kg of dry matter—is comparable to that of concentrated feed sources such as fishmeal, and their digestibility is high, reaching up to 95% in poultry and fish, making them an efficient feed component (Nekrasov *et al.*, 2019).

Inclusion of insect larvae in animal feed improves growth performance and health status. For example,

broiler chickens (Ross 308) fed with diets containing insect protein showed improved feed conversion and reduced pathogenic bacterial load. The inclusion of black soldier fly larvae also enhanced growth rates and positively influenced the gut microbiota (Nekrasov *et al.*, 2019).

Despite their high nutritional value, the use of insect larvae in feed has certain limitations. Until 2021, several countries had restricted the use of insects in livestock feed. Furthermore, the potential accumulation of toxic elements (Ba, Hg, and Mo) in larvae necessitates strict substrate control (Bosch *et al.*, 2014). Negative consumer perception regarding the use of insects in animal feed may also limit their adoption.

Stages of Organic Waste Bioconversion Using *Hermetia illucens*

Preparation of the organic substrate

In recent years, innovative approaches for processing organic waste using insect larvae have received increasing attention. The black soldier fly (*Hermetia illucens*) is one of the most effective agents for bioconversion. The larvae of this species are regarded

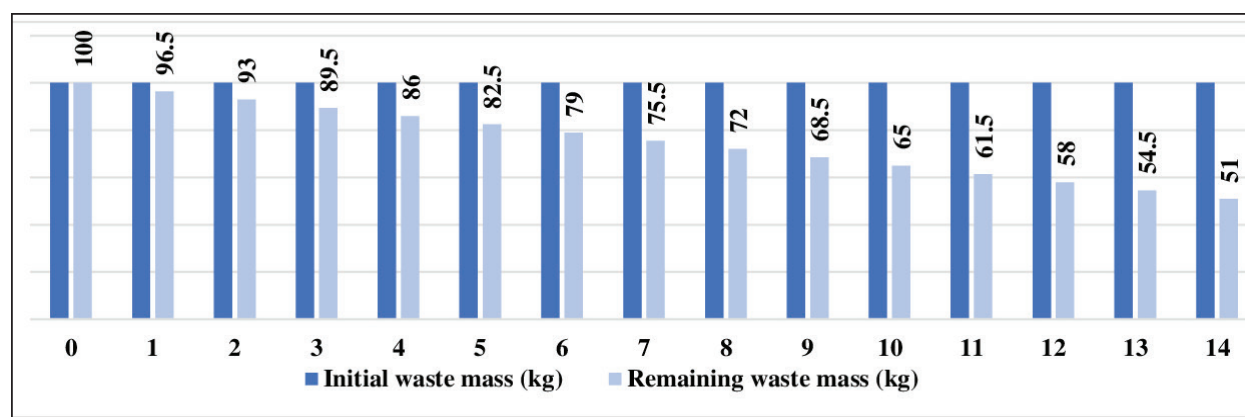


Fig. 4. Reduction in waste mass by days of larval growth (Dewi *et al.*, 2024).

Table 6. Recommended rearing conditions for *Hermetia illucens* Larvae (Diener *et al.*, 2011; Surendra *et al.*, 2016).

Rearing parameter	Recommended value
Temperature	27 °C–30 °C
Air humidity	60%–80%
Substrate moisture	60%–70%
Larval density	5,000–10,000 larvae/kg substrate
Feeding phase duration	7–14 days
Waste mass reduction	50%–80%
Bioconversion rate (larval biomass yield)	15%–25%

as a promising source of feed ingredients due to their rapid growth rate, ability to feed on a wide range of organic substrates, and capacity to produce a valuable protein- and fat-rich biomass (Van Huis *et al.*, 2013; Makkar *et al.*, 2014).

The bioconversion process using *H. illucens* larvae involves several key stages: preparation and loading of organic waste, larval inoculation, substrate decomposition, biomass harvesting, and further processing into finished products (Diener *et al.*, 2009). The first stage involves the collection of organic waste, such as food residues, fruits and vegetables, agro-industrial waste, and by-products from food processing. The waste is shredded, homogenized, and adjusted for moisture content—ideally between 60% and 70%—if necessary. This preparation ensures uniform access to nutrients and accelerates the feeding process. However, the substrate must not contain toxic compounds, antibiotics, or heavy metals, as these can inhibit larval development and compromise the safety of the final product (Barragán-Fonseca *et al.*, 2024).

Figure 4 illustrates the reduction in organic waste mass over time due to the activity of *H. illucens* larvae. The initial waste mass is 100 kg. By day 10 of bioconversion,

less than 60 kg remains, and approximately 50 kg remains by day 14 (Dewi *et al.*, 2024). This demonstrates the larvae's ability to reduce more than 50% of the organic material, significantly decreasing waste volume and contributing to environmentally sustainable waste management (Fitriana *et al.*, 2021).

Inoculation of *H. illucens* larvae and initial growth

Inoculation is typically performed using larvae that are 3–5 days old. On average, between 5,000 and 10,000 larvae are introduced per 1 kg of substrate (Okpoko *et al.*, 2024). The larvae quickly penetrate the substrate, thereby initiating the consumption of organic matter and subsequent growth (Diener, 2009). Optimal development occurs under the following conditions: temperature of 27°C–30°C, relative humidity of 60%–80% (Table 6), and darkness (Wang and Shelomi, 2017).

The average feeding phase lasts between 7 and 14 days, during which the larval mass can increase by a factor of 1,000–1,500 (Barragán-Fonseca *et al.*, 2017). This stage is characterized by the intensive degradation of organic matter and the production of a by-product known as *frass*, which can be used as an organic fertilizer.

Figure 5 illustrates the exponential increase in the mass of *H. illucens* larvae during a 14-day growth period. The larval weight increased from 0.01 to 7.0 g, confirming their high productivity under optimal conditions (temperature: 27°C–30°C; humidity: 60%–70%) (Huang *et al.*, 2025). The most intensive weight gain occurred between days 5 and 11, after which the growth rate stabilized. This dynamic allows for accurate scheduling of larval harvest to maximize biomass yield (Diener *et al.*, 2009).

Harvesting of *H. illucens* Larvae and biomass processing

When larvae reach the prepupal stage, they migrate out of the substrate in search of dry places for pupation, which greatly simplifies collection. The harvested larvae were dried at 60°C–80°C, followed by milling into larval meal. Fat extraction can be performed

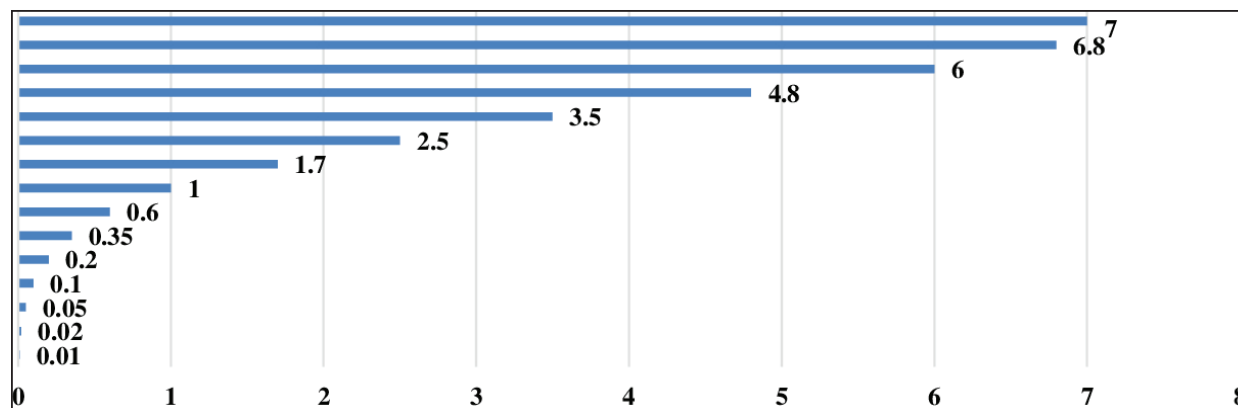


Fig. 5. Larval mass at different growth days (g) (Diener *et al.*, 2009).

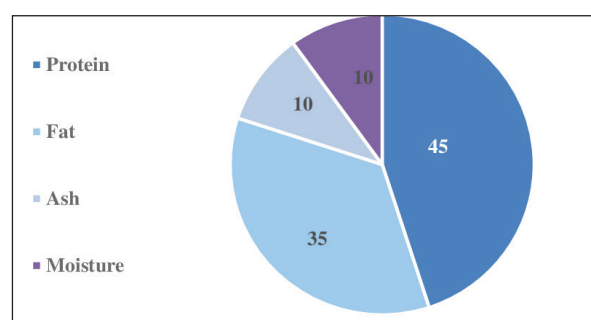


Fig. 6. Chemical composition of dried *Hermetia illucens* larvae (%) (Liu *et al.*, 2017).

using a mechanical press or CO₂ extraction method (Dortmans *et al.*, 2017) if required.

The primary products obtained include: *larval meal* containing 40%–60% protein and 5%–10% fat and *larval oil* rich in lauric acid, which is used as a feed-grade or technical fat (Nyakeri *et al.*, 2017).

After drying, the chemical composition of *H. illucens* larvae is dominated by protein (~45%) and fat (~35%), making them a highly valuable raw material for compound feed. Ash and residual moisture each account for approximately 10% (Fig. 6). *Hermetia illucens* larvae represent a sustainable and efficient alternative to fishmeal and soybean meal in animal diets due to their high protein and lipid content (Liu *et al.*, 2017).

According to Traksele *et al.* (2021), the protein digestibility of *H. illucens* larvae in poultry and fish feed exceeds 85%, while the fats exhibit antimicrobial activity, making them particularly valuable in the diets of animals sensitive to pathogens.

Figure 7 shows the quantities of products obtained from processing 100 kg of organic waste using *H. illucens* larvae. On average, approximately 15 kg of dried larvae is produced, containing 5 kg of fat and 7-kg protein. Additionally, approximately 30-kg frass (organic fertilizer) is generated. These results demonstrate the

effectiveness of this technology in reducing waste and producing valuable feed and agricultural products.

Feed efficiency and applications of *Hermetia illucens*-based products

The resulting protein meal and fats can be used as components in compound feeds for broilers, laying hens, pigs, fish, shrimp, and even pets. The larvae supply animals with high-quality animal protein, B-group vitamins, minerals (Ca, P, Mg), and fatty acids—including lauric acid (C12:0), which has pronounced antimicrobial activity. The use of *H. illucens*-based feed allows the following: to replace up to 100% of fish meal in aquaculture diets (Henry *et al.*, 2015); improve feed conversion ratios; reduce CO₂ emissions and resource consumption. Bioconversion of organic waste using *H. illucens* larvae is a sustainable, economically viable, and environmentally friendly technology. It allows for efficient biomass recycling, waste volume reduction, and high-quality feed ingredient production suitable for industrial-scale use (Table 7).

Organic resources for bioconversion: suitable substrates for *Hermetia illucens*

Bioconversion of organic waste using *larvae of H. illucens* (black soldier fly) is becoming an increasingly popular technology for sustainable biomass management and feed production. The process efficiency largely depends on the quality and composition of the initial raw materials.

Food waste is one of the most common and accessible types of raw materials for bioconversion. It includes food scraps from restaurants, cafeterias, and supermarkets, as well as vegetable and fruit peels (Diener *et al.*, 2011; Van Huis *et al.*, 2013). Food waste is characterized by high moisture and nutrient content, which promotes rapid growth of larvae.

Agricultural residues—such as straw, crop tops, grain processing byproducts, and vegetable and fruit peels—are also used in bioconversion (Makkar *et al.*, 2014; Surendra *et al.*, 2016). To enhance nutrient accessibility, these materials often require pretreatment (e.g., grinding or moistening).

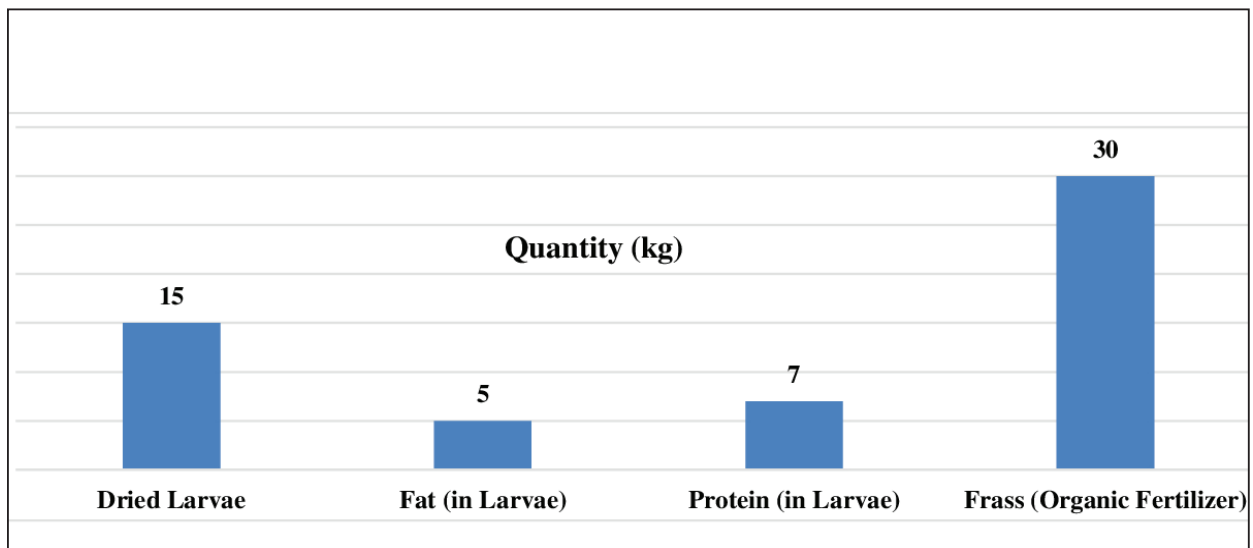


Fig. 7. The quantities of products obtained from processing 100 kg of organic waste using *H. illucens* larvae.

Table 7. Advantages of using *Hermetia illucens* Larvae in animal feeding (Henry *et al.*, 2015).

Advantages of using Larvae in feeding	Description
High nutritional value	40%–60% protein, all essential amino acids
Alternative to fishmeal and soybean meal	Partial or full replacement
Environmental sustainability	Reduced environmental impact
Bio-protective effect	Contains antimicrobial fatty acids
Flexibility in use	Applicable in various livestock sectors
Economic efficiency	Lower raw material costs, potential for local production

Animal-derived waste can be used provided that it strictly complies with sanitary regulations. This includes meat and fish byproducts, poultry manure, and other farm waste (Nguyen *et al.*, 2015; Jucker *et al.*, 2017). However, such waste requires careful monitoring due to the risk of pathogenic microorganisms.

Household food waste and industrial food processing byproducts, including dairy residues and brewery waste, are also recognized as promising raw materials for bioconversion (Li *et al.*, 2011; Lalander *et al.*, 2019).

Conversely, certain types of waste, such as inorganic materials, toxic, or chemically contaminated substrates, are unsuitable for bioconversion by larvae due to the risk of contaminating the final product and reducing processing efficiency (Gold *et al.*, 2020).

Thus, selecting the optimal type of organic waste is a critical factor for the successful application of *H. illucens* larvae bioconversion technology to produce feed and reduce environmental impact.

Factors affecting the efficiency of bioconversion of organic waste by *Hermetia illucens* Larvae

The efficiency of organic waste bioconversion using larvae of *H. illucens* (black soldier fly) depends on a

combination of biotechnological and environmental factors that determine larval growth rate, substrate utilization efficiency, and the quality of the final product (Surendra *et al.*, 2016; Lalander *et al.*, 2019). Temperature is a critical parameter that affects larval metabolism and development. The optimal temperature range for rearing *H. illucens* is approximately 27°C–30°C. Lower temperatures slow growth rates, whereas temperatures above 35°C cause stress and increased larval mortality (Diener *et al.*, 2011; Nguyen *et al.*, 2015). A stable temperature ensures maximum bioconversion efficiency and biomass quality.

Substrate moisture should be maintained within 60%–70% to provide larvae with optimal feeding conditions. Substrates with low moisture content limit nutrient accessibility and reduce larval activity, whereas excessive moisture promotes pathogen growth and anaerobic processes (Jucker *et al.*, 2017).

The chemical composition and physical properties of the substrate significantly influence the bioconversion efficiency. Protein- and easily digestible carbohydrate-rich substrates promote rapid larval growth and higher biomass yield (Makkar *et al.*, 2014). To enhance nutrient availability, fibrous or tough materials require

pretreatment (e.g., grinding or fermentation) (Surendra *et al.*, 2016).

Larval stocking density: The optimal stocking density varies depending on the substrate type and quantity but typically ranges from 5,000 to 10,000 larvae per 1 kg of substrate. Overly high densities lead to food competition, slowing growth, and increasing mortality, whereas too low a density reduces system productivity (Diener *et al.*, 2011; Lalander *et al.*, 2019).

Aeration and environmental conditions: adequate oxygen supply and proper ventilation prevent the accumulation of harmful gases (e.g., ammonia) and the growth of anaerobic microorganisms, improving larval growth conditions and reducing disease risks (Gold *et al.*, 2020).

Rearing duration: The optimal growth phase typically lasts 10–14 days. Harvesting too early results in low biomass yield, whereas overly prolonged rearing leads to substrate overuse and reduced larval quality (Nguyen *et al.*, 2015).

Effective management of these factors significantly enhances the efficiency of bioconversion using *H. illucens* larvae, reducing waste volume, and producing high-quality feed material. Optimizing rearing parameters for each specific waste type is recommended to maximize productivity.

Global Approaches and Innovations in Bioconversion of Waste into Feed

Global spread of bioconversion technology involving *H. illucens*

Bioconversion technology using black soldier fly larvae (*Hermetia illucens*) has gained widespread adoption across many regions worldwide due to its high efficiency in processing various organic wastes. Infrastructure for specialized farms cultivating *H. illucens* larvae is currently developing in Europe, Asia, North America, and Africa. These enterprises create new value-added chains within the agro-industrial complex by combining waste management solutions with the production of high-quality feed ingredients (Li *et al.*, 2011; Lalander *et al.*, 2019).

The main directions of technology implementation include the following:

- **Food waste utilization:** The technology enables significant reduction of biowaste volumes at food processing plants, supermarkets, and restaurants while simultaneously producing high-protein feedstock. This approach reduces landfill burdens and lowers GHG emissions (Makkar *et al.*, 2014).
- **Feed production for livestock and aquaculture:** *H. illucens* larvae biomass is used as an ingredient in compound feeds for poultry, pigs, fish, and other animals, reducing dependence on traditional protein sources such as fishmeal and soy, with economic and environmental benefits (Jucker *et al.*, 2017).
- **The development of a circular economy.** Organic waste bioconversion is integrated into closed

production loops where waste is converted into valuable resources. After larval processing, residues are used as organic fertilizers, promoting sustainable agriculture and reducing environmental pollution (Gold *et al.*, 2020).

- **Scientific and technical innovations.** The current research focuses on optimizing larval rearing conditions, improving feed product quality, increasing bioconversion efficiency, and scaling production processes. Simultaneously, regulatory authorities in several countries are developing frameworks to ensure the safety and standardization of bioconversion products used in feed production (EFSA, 2015, 2021).

Overall, the adoption of *H. illucens* larvae in organic waste management and feed production systems reflects a global trend toward sustainable development. This technology addresses critical challenges related to food security and environmental responsibility by combining innovative biotechnology with the principles of circular economy.

Waste bioconversion using *Hermetia illucens* Larvae in different regions of the world

The development of bioconversion technology using black soldier fly (*Hermetia illucens*) larvae is advancing rapidly across various continents due to its high ecological and economic efficiency (Opoku *et al.*, 2023). The successful integration of this technology into waste management systems and feed production is illustrated in the following examples.

European Union (Netherlands, France, and Germany). The Dutch company *Protix* is considered one of the global leaders in insect-based protein production. It has successfully implemented *H. illucens* larvae rearing technologies for processing food waste and produces feed lines for pets, fish, and poultry. Their facility processes thousands of tons of organic waste annually, simultaneously producing high-quality protein and organic fertilizers (Van Huis *et al.*, 2013; Gold *et al.*, 2018).

Ynsect, a French company, is also developing large industrial complexes focused on producing insect-based feed protein and fertilizers, investing heavily in research and sustainable technologies.

Asia (China, South Korea, and Vietnam). In China, a farm in Guangdong Province processes over 50 tons of food and kitchen waste daily using *H. illucens*, producing both protein feed and fat for use in pig and poultry diets (Yu *et al.*, 2020).

According to Nguyen *et al.* (2015), the use of *H. illucens* larvae has become an important solution for agricultural waste management in Vietnam and provides local farmers with sustainable feed alternatives.

As part of a national strategy to reduce biowaste in urban and suburban areas, South Korea is promoting government initiatives to implement insect-based technologies (Surendra *et al.*, 2016).

Africa (Kenya, Uganda, and South Africa). The technology has gained widespread popularity in Kenya, Uganda, and other African countries due to its accessibility and ability to address feed shortages. The Kenyan startup *InsectiPro* collaborates with smallholder farmers to train them in the production of *H. illucens* larvae for use in poultry and fish feed, which contributes to food security and improves the profitability of rural farms (Lalander *et al.*, 2019).

In South Africa, the company *AgriProtein* is a pioneer in commercial insect-based waste processing. Its facilities process up to 250 tons of organic waste per day, providing sustainable protein sources for the feed industry. *Latin America (Brazil and Colombia)*. In Brazil, several projects focus on processing sugarcane and corn residues. Agricultural universities are conducting experiments to integrate *H. illucens* into organic waste recycling cycles. As part of a national sustainable agriculture strategy, Colombia actively employs bioconversion technologies to produce protein and organic fertilizers. *North America (USA and Canada)*. The American company *Enterra Feed Corporation* (also operating in Canada) specializes in the production of protein feeds and fertilizers from *H. illucens*. Their facilities process food and agricultural waste, creating a circular production model that minimizes waste (Jucker *et al.*, 2017).

In the United States, ongoing research is exploring the integration of insect-derived products into aquaculture feed chains to reduce pressure on wild fish stocks.

These examples from various regions demonstrate the high potential of waste bioconversion using *H. illucens* larvae as a tool for sustainable development. They confirm that integrating larvae into agri-food and environmental systems can yield both economic benefits and a significant reduction in ecological impact regardless of geography.

Industrial bioconversion of organic waste using *Hermetia illucens*

In recent years, significant technological advancements have emerged in the field of organic waste bioconversion, particularly aimed at the industrial scaling of processes involving the black soldier fly (*Hermetia illucens*). These innovations encompass the automation of production cycles and the enhancement of engineering solutions for managing microclimate, substrate, and insect development stages.

The development of automated larval rearing systems, which include controlled modules for substrate delivery, ventilation, temperature, and humidity regulation, is a key technological breakthrough. For instance, “containerized Black Soldier Fly Larvae production units” enable the bioconversion cycle to be conducted in closed climate-controlled environments with a high level of biosecurity, allowing such facilities to be located even within urban areas (Pahmeyer *et al.*, 2020).

In addition, smart monitoring and control platforms based on the internet of things and machine learning

have been introduced. These systems track larval behavior, substrate consumption, and growth rates in real time, enabling prompt environmental adjustments to improve productivity and reduce processing time (Avila *et al.*, 2022).

Another significant achievement is the optimization of logistics and engineering solutions for sorting and drying larval biomass, as well as for the disposal of residues (*frass* fractions). Modern installations can produce standardized protein and fat feed ingredients that comply with international safety standards (Van Huis, 2020; Barragan-Fonseca *et al.*, 2021).

Large-scale farms, such as Ynsect (France), Protix (Netherlands), and AgriProtein (South Africa), have become pioneers in the commercialization of bioconversion technologies and have demonstrated successful industrial-scale operations, processing tens of tons of waste daily (Byrne, 2024).

Thus, technological advancements in waste bioconversion using *H. illucens* are focused on creating highly efficient, sustainable, and safe solutions capable of providing alternative feed sources while simultaneously reducing environmental impact.

Regulatory framework and safety assurance for *Hermetia illucens* Larvae-derived feed products

With the advancement of waste bioconversion technologies using *H. illucens* larvae, compliance with regulatory requirements and product safety assurance have become critical factors for the sustainable integration of these solutions into the feed industry. Regulatory authorities worldwide are increasingly developing legal frameworks aimed at controlling substrates, processing conditions, and final product quality (Regulation (EU) 2017/893) (Table 8).

Use of black soldier fly Larvae (*Hermetia illucens*) as a feed component requires strict compliance with biological and chemical safety standards

The cultivation, processing, and storage of *H. illucens* larvae for use in animal feed must adhere to stringent safety protocols to ensure both biological and chemical safety. The following key factors should be considered:

- **Microbiological safety:** Hygienic conditions during larval rearing and processing are crucial for preventing feed contamination with pathogenic microorganisms. Controlling pathogens such as *Salmonella spp.*, *Escherichia coli*, *Listeria monocytogenes*, and other potentially harmful bacteria According to EFSA (2021a), maintaining thermal treatment conditions of at least 90 °C for 15 minutes effectively reduces microbiological risks.

- **Chemical safety (Heavy Metals and Toxins):** One of the critical concerns is the absorption of heavy metals (lead, cadmium, mercury) and mycotoxins from substrates. Studies have shown that the concentration of toxic substances in larvae can directly depend on the composition of the substrate used; therefore, prior chemical analysis is required for the application of organic waste (Van der Fels-Klerx *et al.*, 2018).

Table 8. Key aspects of regulation and safety.

Country / Region	Permitted substrates	Approved use areas	Main safety restrictions	Regulatory authority
European Union	Plant waste, food industry by-products	Aquaculture, pig farming, poultry	Ban on manure, household waste; strict microbiological control	Regulation, EU 2017/893
USA	Plant-based food waste	Feed for poultry, fish, pets	No pathogens or heavy metals; compliance with AAFCO standards	FDA, 2024 AAFCO, 2024
China	Food waste, plant substrates	Poultry, pig farming	Under development; microbiological monitoring in place	Ministry of Agriculture and Rural Affairs, Yangyang Li et al., 2018
African Countries (e.g., Kenya, South Africa)	Locally sourced organic waste (quality controlled)	Feed for poultry, fish, pigs	Introduction of national hygiene standards	FAO, 2012 National agencies support

Residues of antibiotics and pharmaceuticals: When using food or agricultural waste, it is important to eliminate the presence of antibiotic residues, hormones, and pesticides, as they may accumulate in the larvae and enter the food chain. EU standards (Regulation (EC) No 767/2009) and AAFCO (USA) require the mandatory testing of residual substances.

Allergens: Although allergic reactions to insects are rare, potential cross-reactivity with crustaceans and mites has been reported. Therefore, in Europe, *H. illucens*-derived feed products must be appropriately labeled (EFSA, 2021b).

Processing and preservation: To minimize risks and extend shelf life, larvae should undergo thermal processing (e.g., drying, boiling, roasting) and be stored under controlled conditions (humidity, temperature, pest exclusion) (EFSA, 2015).

Traceability and raw material control: Each batch of larvae and substrates used must be accompanied by documentation verifying the origin, rearing conditions, and laboratory test results, which is essential for supply chain traceability and quality control.

Research by EFSA (2021a, b), Schlüter et al. (2017), and Van der Fels-Klerx et al. (2018) confirmed that when rearing and sanitation standards are followed, *H. illucens* larvae represent a safe and reliable protein source for feed applications.

Feed Solutions Based on *Hermetia illucens* Larvae: Application and Effectiveness Areas

Use of *H. illucens* larvae in animal feed

In recent years, black soldier fly (*Hermetia illucens*) larvae have been increasingly integrated into feed production as a promising source of high-quality protein and fat. Their ability to efficiently convert organic waste, combined with rapid growth and adaptability, makes them particularly suitable for use in poultry farming, aquaculture, and pig production (Henry et al., 2015; Barragán-Fonseca et al., 2017; Schiavone et al., 2017; Parodi et al., 2018; Elahi Usman et al.

2022). Larval feed contains up to 42%–45% crude protein and up to 35% fat, as well as antimicrobial and immunostimulatory compounds such as lauric acid and chitin (Mwaniki et al., 2018; Biasato et al., 2019).

Studies have shown that replacing up to 15% of soybean or fishmeal with *H. illucens* meal in broiler and layer diets does not negatively affect productivity and often improves weight gain, feed conversion ratio, and meat and egg quality. Positive changes in gut microbiota composition and enhanced immune function have also been observed (Mwaniki et al., 2018).

In aquaculture, insect meal has been successfully used in feeds for tilapia, trout, catfish, carp, and shrimp, as it can replace up to 50% of fishmeal without compromising growth performance, with additional benefits for gut health and disease resistance (Henry et al., 2015; Van Huis, 2020).

Introducing 5%–10% insect meal into starter diets for piglets shows positive effects on growth, gastrointestinal health, and immunity. However, at levels above 15%, digestibility may decrease due to chitin content, indicating the need for additional processing steps (Biasato et al., 2019).

Different inclusion levels of *H. illucens* meal in animal diets affect productive and physiological parameters in various ways. Moderate levels (up to 10%–15%) generally lead to positive outcomes, whereas higher levels (>20%) may reduce digestibility, particularly in young animals, due to the higher chitin content (Table 9).

The final product's effectiveness and safety depend on adherence to technological parameters and hygiene standards. Many European producers are implementing the HACCP and ISO 22000 standards in the insect processing stage.

Table 9. Nutritional value of *Hermetia illucens* Larvae meal compared to conventional protein sources (Barragán-Fonseca *et al.* (2017); Henry *et al.* (2015); Gasco *et al.* (2020).

Parameter	<i>H. illucens</i> Larvae meal	Fishmeal	Soybean meal
Crude protein (%)	42–45	60–72	44–50
Crude fat (%)	25–35	9–12	1–2
Lauric acid (%)	35–40 of total fat	Absent	Absent
Chitin (%)	6–8	<0.5	None
Lysine (%)	2.7	4.5	3.0
Met + Cys (%)	1.1	2.2	1.4
Calcium (%)	0.6–1.5	4–6	0.2
Phosphorus (%)	0.5–1.3	2.5–3.5	0.6–0.7

Table 10. Comparative nutritional value of various additives (per 100 g dry matter) (Makkar *et al.*, 2014; Henry *et al.*, 2015).

Parameter	<i>Hermetia illucens</i>	<i>Spirulina spp.</i>	Soybean meal	Fishmeal
Crude protein (%)	42–60	55–70	44–48	60–72
Crude fat (%)	15–35	6–8	2–4	8–12
Lysine (%)	2.5–3.0	3.2	2.8	5.1
Met + Cys (%)	1.6–2.0	2.1	1.2	2.3
Calcium (%)	5.0–7.0	0.3	0.25	4.2
Phosphorus (%)	1.5–2.0	1.1	0.7	3.0

Nutritional advantages and application results of *Hermetia illucens* Larvae in feeding terrestrial and aquatic animals

Improving the efficiency of animal and aquaculture feeding has become a key objective in the context of global population growth and increasing demand for animal-derived products. A special focus is given to alternative protein sources, environmentally sustainable feed additives, and functional ingredients. Among these are insect proteins (*Hermetia illucens*), fermented products, algae, probiotics, and other innovative additives that demonstrate high nutritional value and positive effects on animal productivity.

High levels of crude protein, lipids, amino acids, minerals, and bioactive compounds characterize alternative feed additives (Table 10). The addition of insect protein and spirulina improves growth and egg quality (Tables 11 and 12).

Innovative feed ingredients (insect proteins, microalgae, probiotics, etc.) not only provide a high nutritional value but also significantly improve productivity and resilience in livestock and aquaculture species. These additives play a crucial role in the transition toward more sustainable, profitable, and environmentally friendly systems in animal husbandry and aquaculture.

Nutritional value of *Hermetia illucens* Larvae meal and traditional feed ingredients

Fishmeal remains the leader in protein content, but *Hermetia illucens* (BSFL) offers a more environmentally friendly production process and

a well-balanced amino acid profile. Fishmeal may contain heavy metals and toxins, whereas soybean meal often contains pesticide residues (Weththasinghe *et al.*, 2022). Properly processed BSFL meal poses minimal risks. Moreover, larvae cultivation contributes to organic waste recycling and requires significantly fewer resources than fish or soybean production (Fatima *et al.*, 2023).

While soybean and fishmeal are currently more affordable and well-established on the market, advancements in insect production technologies are making BSFL meal increasingly cost-competitive (Table 13).

Problems and Limitations in the Use of *Hermetia illucens* Products

Technical challenges in optimizing bioconversion

Despite the significant advantages of waste bioconversion technologies using insects—particularly *H. illucens* larvae—the practical implementation of this innovation faces several serious challenges and limitations. These concerns technical, economic, regulatory, and environmental aspects.

Bioconversion processes encounter several technical obstacles. The *variability in the composition of organic waste* complicates the standardization of processing parameters, as different types of waste require individualized treatment conditions (Table 14).

One of the main technical challenges is the *high variability in the nutrient composition, moisture*

Table 11. Use of *Hermetia illucens* Larval meal in livestock and aquaculture.

Animal Species	<i>Hermetia illucens</i> application	Soybean application	Fishmeal application	Authors
Poultry (layers, broilers)	Source of protein and fat; enhances immunity and product quality	Primary protein source	Feed additive; improves growth and meat quality	Zhu et al., 2022; Yu et al., 2019; Wang, 2017
Pigs	Well-digested; replaces soybean protein	Main plant-based protein	Rarely used due to high cost	Zhu et al., 2022; Go et al., 2022; Boontiam & Khaodhiar, 2022
Fish & Aquaculture	Suitable as partial fishmeal replacement	Limited use	Main protein source	Mohan, 2022; Ling, 2024; Vodounnou, 2025
Cattle	Used in compound feed; stimulates rumen microflora	Widely used	Rarely used	Schmitt, 2020; Ogello, 2025

Table 12. Effect of insect protein in pigs (Zhu et al., 2022).

Items	C	T1	T2	p-value
Initial body weight (kg)	7.82 ± 0.23	7.43 ± 0.37	7.62 ± 0.29	0.597
Final body weight (kg)	63.98 ± 2.53	60.94 ± 2.70	66.38 ± 3.51	0.427
Average daily gain (kg/d)	0.47 ± 0.01 ^b	0.46 ± 0.02 ^b	0.56 ± 0.03 ^a	0.029

Data are means for 8 replicates of 1 pig/replicate. Abbreviation: C, control group; T1, supplement 4% black soldier fly; T2, supplement 8% black soldier fly. All data are expressed as mean ± standard error. All data were tested for significance using one-way ANOVA and Tukey's post hoc test. Differences were identified as significant at $p < 0.05$. ^{a,b} Bars with different letters are different among diets with black soldier fly treatments.

Table 13. Comparison of *Hermetia illucens* Larvae with traditional feeds (Soy and Fishmeal) (Supriyatna et al., 2018).

Indicator	<i>Hermetia illucens</i> Larvae	Soybean meal	Fishmeal
Protein content	40%–60% (depending on stage and processing)	40%–48%	60%–72%
Fat content	20%–35%	1.5%–3%	8%–12%
Amino acid profile	Balanced, rich in lysine and methionine	Rich in lysine, low in methionine	Excellent profile
Digestibility	High, especially after processing	Good	Very high

Table 14. Key limitations of bioconversion technology using insects.

Category	Problems and limitations	Authors
Technical	Substrate variability, scalability issues, high sensitivity to environmental conditions	Tariq, et al., 2025 Siddiqui et al., 2025
Economic	High capital costs, elevated production cost, low profitability	Ayompe, 2025
Regulatory	Legal restrictions, lack of harmonized standards, consumer skepticism	Tariq, et al., 2025
Environmental	Greenhouse gas emissions, challenges in frass management, risk of environmental pollution	Lisboa et al., 2024

content, and potential contamination of waste substrates, which can significantly affect larval growth performance and final product stability (Makkar et al., 2014).

Furthermore, scalability remains a critical issue. Under industrial conditions, it is difficult to ensure uniform substrate distribution, temperature and humidity control, and adequate ventilation—all of which are

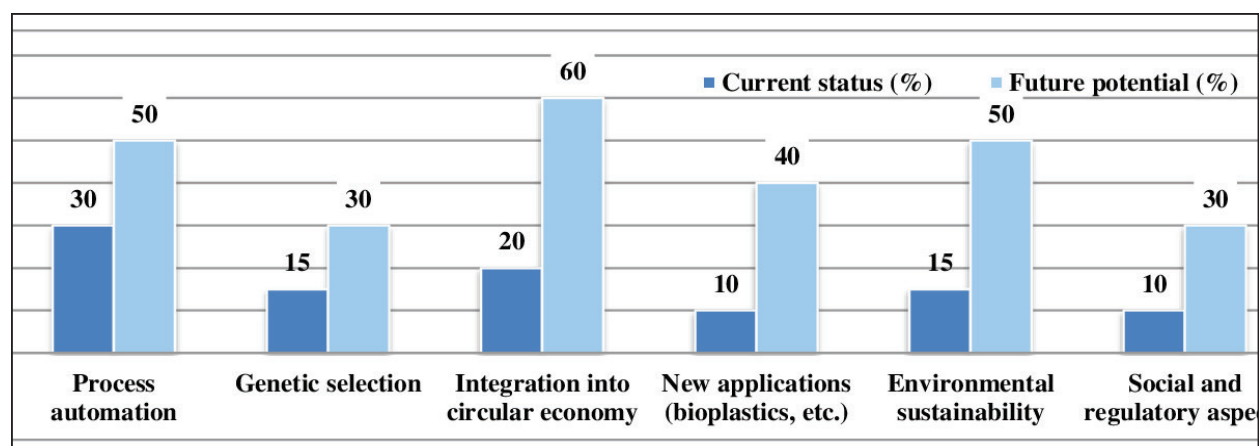


Fig. 8. Current versus future research direction in *Hermetia illucens* Bioconversions (by Wang *et al.*, 2025).

essential for maintaining insect health and productivity (Van Huis, 2013).

In addition, technical difficulties arise in *automating sorting, drying, and processing operations*, which often require expensive engineering solutions and innovative technologies (Wang and Shelomi, 2017).

Economic barriers

Insect-based feed production is relatively expensive due to high initial investments in equipment, climate control systems, and infrastructure for handling and storing organic materials (Van Huis and Oonincx, 2017). The lack of developed logistics for the collection and transport of organic waste is an additional burden. Moreover, the *final cost of the product often exceeds that of conventional protein feed ingredients*, such as soybean meal or fishmeal, hampering broader market acceptance. There is also *limited understanding of the economic viability* of such projects among investors and farmers, especially in countries with unstable economies or where governmental support is lacking (Delivering insect-based feed at scale. 2023).

Regulatory and consumer perceptions

At the international level, no *unified regulatory framework* governing the use of insects in feed production exists. Within the European Union, insect-based feeds are approved only for certain animal categories and only when they are produced from safe substrates (EFSA, 2021a).

Environmental considerations

Although the bioconversion of organic waste using insects is considered an environmentally sustainable technology, *certain ecological risks remain*. Under intensive production conditions, *greenhouse gas emissions*—including methane and nitrous oxide—may occur, especially when moisture levels are imbalanced or substrates begin to decompose (Smetana *et al.*, 2019a).

The management of residual frass—a by-product remaining after larval rearing—is another environmental challenge. Although frass has potential

as a biofertilizer, its properties and safety heavily depend on the initial substrate and processing methods. Without proper control, it may become contaminated with pathogens or heavy metals (Helen *et al.*, 2024).

Future Potential and Research Directions

Innovations in bioconversion technologies

The advancement of *bioconversion technology based on H. illucens* offers *promising opportunities for agriculture and related sectors*. Current research and development efforts focus on optimizing production processes, expanding application areas, and enhancing system sustainability within the circular economy framework.

The automation and digitalization of bioconversion processes is one of the key directions for future innovation. The use of sensors, environmental monitoring systems, and AI algorithms will increase the efficiency of waste-to-biomass conversion, minimize losses, and reduce production costs (Barragán-Fonseca *et al.*, 2021).

Genetic selection and breeding of larvae with improved traits, such as faster growth rates, higher feed conversion efficiency, and resistance to pathogens or variable environmental conditions, is another promising area. Modern biotechnological tools, including genome editing and microbiome analysis, support a deeper understanding of larval physiology and improve it.

Modern biotechnological approaches, including genome editing and metagenomic analysis of the microbiome, contribute to a deeper understanding of insect physiology and its potential for regulation (Xiong *et al.*, 2024).

Figure 8 illustrates a comparative assessment of the current state and projected future potential of key research directions related to the bioconversion process using *H. illucens* (black soldier fly).

Integration into the circular economy (60%) and environmental sustainability (50%) have the highest

Table 15. Promising research directions in the field of bioconversion using *Hermetia illucens* (Amoshie et. al., 2024; Chong et al., 2024; Su et al., 2025).

Direction	Examples of specific research
Automation and technologies	Sensors, AI, climate control, robotic larva sorting
Genetics and breeding	Breeding fast-growing lines, microbiome studies
Circular economy	Integration into farms, manure and feed residue processing
New application sectors	Bioplastics, fertilizers, biofuels, pharmaceuticals
Environmental sustainability	GHG emissions analysis, residue monitoring, life cycle assessment
Social and Regulatory Aspects	Consumer perception, development of standards and guidelines

future potential, reflecting a strong global emphasis on sustainable food systems and waste valorization.

Although *process automation* has made some progress (30%), it remains a top priority for future development (50%), especially in industrial-scale operations.

Genetic selection is still in its early stages (15%), but its importance is growing (30%) as the demand for optimized larval strains increases.

New applications and *social/regulatory aspects* currently receive less attention (10%), yet they are expected to become more relevant (30%–40%) as the market matures.

Overall, Figure 8 highlights the *growing interest in scalable, eco-friendly, and multifunctional insect bioconversion systems* and suggests clear directions for further academic and industrial investment.

Integration into circular economy models

The cultivation technology of *H. illucens* fits perfectly into the concept of the circular economy, where the waste of one industry becomes a resource for another. Examples include: utilization of food and agricultural waste, the return of frass as a biofertilizer for crop production, and the incorporation of the protein and lipid fractions of larvae into feed production chains and even pharmaceuticals (Surendra et al., 2016; Mina et al., 2023; Lu et al., 2025).

Moreover, insect-rearing installations can be integrated into agro-industrial complexes, where bioconversion is a tool for processing internal organic waste streams (e.g., manure and feed residues) into economically valuable resources.

Application expansion across industries

Hermetia illucens biomass and its derived products can be applied not only in the feed industry but also in other promising sectors: *bioplastics* (e.g., based on chitin and proteins from larval exoskeletons) (Vital-Vilchis and Karunakaran, 2025); *biofuels* (larval fat can be converted into biodiesel) (Mohan et al. 2023); *cosmetics and pharmaceuticals* (larval extracts possess antibacterial and antioxidant properties) (Pimchan et al., 2024); and *fertilizers and plant growth stimulators* based on *frass* enriched with microbiota and amino acid residues (Fuhrmann et al., 2022).

Research gaps and future directions

Despite significant progress, substantial gaps remain in the scientific and practical foundation that require further exploration: *long-term environmental impacts* of large-scale insect farming (e.g., greenhouse gas emissions, effects on biodiversity), the *safety and quality* of insect-based products depending on the substrate types used, the *optimization of nutritional value* and bioavailability of proteins, lipids, and micronutrients for various species of farm animals and fish, the *impact on animal health and productivity* when feeds containing insect-derived ingredients are consumed, and the *social perception* and factors influencing the acceptance of this new technology by consumers and producers (Table 15). Table 15 presents an overview of the most promising bioconversion research directions using *H. illucens*. Key areas such as automation, genetics, integration into CES, and emerging applications such as bioplastics and biofuels are highlighted. These directions reflect the growing scientific and commercial interest in optimizing insect-based technologies’ sustainability, efficiency, and versatility.

Conclusion

BSF larvae (*Hermetia illucens*) offer a sustainable and efficient solution for converting diverse organic wastes into protein- and lipid-rich biomass suitable for animal feed, biofertilizers, and other applications. Their high substrate conversion rates and bioactive compound production highlight their potential for advancing circular economy approaches and sustainable protein production. Although progress has been made in optimizing rearing conditions and postharvest processing, challenges remain, including substrate standardization, microbial risk management, cost reduction, and production scaling up. Successful commercialization and broader adoption of BSF-derived products will also depend on supportive regulatory frameworks and social acceptance. Overall, BSF bioconversion represents a promising technology for transforming waste management and animal nutrition, contingent on continued research, industry collaboration, and regulatory development.

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Conflict of interest

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Authors contribution

The author solely contributed to the conception and design of the review, data collection, drafting, and manuscript revision.

Data availability

All data are provided in the revised manuscript.

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